Tanmay Sutar

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APPLIED AI EXPERIENCE

Applied AI Engineer Sep 2025 – Present

Magic Hour (YC W24)

- Built an OpenAI Agents SDK video agent with staged tools, persistent project state, streamed tool traces, and per-artifact provenance, token usage, and cost accounting for long-running multimodal workflows.
- Developed a YouTube grounding pipeline combining retrieval, transcript scoring, visual checks, and script review to align generated narration with source footage; added durable SQLite jobs, bounded retries, and provider recovery.
- Built a footage-grounded human-feedback lab with frozen source evidence, prompt/model fingerprints, validated rewrite annotations, and JSONL exports for controlled prompt evaluation.

RESEARCH EXPERIENCE

Reinforcement Learning for Recommendation Following Sep 2026

BlueDot AI Safety Sprint | Code & results | Write-up

- Trained Qwen2.5-3B with GRPO and rank-8 LoRA on 240 benign prompts; designed a matched correctness-rewarded control and 1,056 evaluation inputs with frozen peer recommendations.
- Measured wrong-advice agreement rising from 13/48 to 48/48 under a single-letter protocol; used prompt ablations and next-token logits to diagnose a brittle copying policy and an accuracy tradeoff (85.4% to 70.8%).

Teacher-Student Fine-Tuning and Behavioral Evaluation Jan 2026 – Sep 2026

BlueDot AI Safety Sprint | Write-up

- Used the Tinker API for LoRA supervised fine-tuning of Llama-3.1-8B teachers on dialogue data; implemented JSONL conversion, checkpoint tracking, and hashed dataset/config manifests.
- Trained students on 25,649 number-only sequences with completion-only loss; audited data and model lineage, rebuilt an invalid control using untouched-base-generated data, and ran inference on Modal/vLLM.
- Evaluated 225 held-out contexts with two automated judges and dialogue-cluster bootstrap analysis: 51/225 unsafe labels versus 0/225 for the clean control; reported judge sensitivity and one-seed, in-domain limits.

Echoes of Human Malice in Agents 2025 – Present

Social Wellbeing Lab, Georgia Tech | Research Assistant (Oct 2026 –) | Co-author | ICWSM 2027 (accepted) | Paper

- Wrote a data-parallel vLLM load balancer (8×H200 + 4×L40S) for 10K ten-turn conversations at 4× throughput.
- Jailbreak-tuned Llama-3.1-8B with SFT and DPO, raising attack success from 57.3–64.2% to 95.8–96.9%.
- Built the multi-turn adversarial benchmark and model-judge eval pipeline for Llama-3.1-8B and Gemini-2.0-Flash.

Refusal Ablation in FLUX 3 Image Klein Oct 2026 – Present

Georgia Tech

- Removing refusals from FLUX 3 Image Klein (5.6B) by orthogonalizing its weights against the refusal direction.

Local Language-Model Personalization Jun 2026

Independent research

- Fine-tuned Gemma on personal writing samples with MLX on Apple Silicon for local writing-style personalization.

Activation Probing of Code Models Jan 2026

Independent research

- Probed Qwen2.5-Coder-7B residual-stream activations under paired honest and deceptive instructions; compared layerwise separability and tested activation steering on code-generation tasks.

EDUCATION

Georgia Institute of Technology Aug 2023 – May 2025

M.S. Computer Science, Machine Learning Specialization

Indian Institute of Technology Guwahati Aug 2019 – May 2023

Bachelor of Technology

TECHNICAL SKILLS

Model training: PyTorch, Transformers, PEFT, TRL, SFT, DPO, GRPO, LoRA/QLoRA, Tinker, MLX

Research & evaluation: Controlled experiments, LLM-as-judge, bootstrap analysis, activation probing, steering, weight orthogonalization

Systems: Data-parallel multi-GPU inference, Python, TypeScript, SQL, OpenAI Agents SDK, vLLM, Modal, SQLite, Docker, Git